LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 11 Test: Electricity

Chapter / Topic: Electricity

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SECTION A

Q1. A wire of resistance R is cut into five equal parts, which are then connected in parallel. If the equivalent resistance is R′, the ratio R/R′ is: **[1]**

Answer: (c) 25. Each part is R/5, and five in parallel give R′ = R/25.

Q2. Which of the following does NOT represent electric power in a circuit? **[1]**

Answer: (d) IR^2

Q3. The SI unit of electric charge is the: **[1]**

Answer: (b) coulomb

Q4. How much current does an 1100 W appliance draw from a 220 V supply? **[1]**

Answer: (b) 5 A. I = P/V = 1100/220

Q5. An electric bulb is rated 220 V, 100 W. When it is operated on 110 V, the power consumed is: **[1]**

Answer: (d) 25 W. $R = V^2/P = 484\ \Omega$, and $P = 110^2/484 = 25$ W

Q6. **Assertion (A):** An ammeter is always connected in series with the component whose current is to be measured. **Reason (R):** The same current flows… **[1]**

Answer: (a)

SECTION B

Q7. How much work is done in moving a charge of 2 C across two points that have a potential difference of 12 V? **[2]**

Answer: W = QV = 2 C × 12 V = 24 J

- Formula: 1
- 24 J: 1

OR, answer: Alloys have a higher resistivity than pure metals, so they produce more heat. They also do not oxidise (burn) easily at high temperatures.

- Higher resistivity: 1
- Do not oxidise easily: 1

Q8. Resistors of 4 Ω and 6 Ω are connected in series to a 12 V battery. Find the current in the circuit and the potential difference across the 4 Ω… **[2]**

Answer: R = 4 + 6 = 10 Ω. I = V/R = 12/10 = 1.2 A. V across 4 Ω = IR = 1.2 × 4 = 4.8 V.

- Current 1.2 A: 1
- 4.8 V: 1

SECTION C

Q9. A copper wire has a diameter of 0.5 mm and a resistivity of $1.6 \times 10^{-8}\ \Omega$ m. **[3]**

(i) $A = \pi d^2/4 = 3.14 \times (0.5 \times 10^{-3})^2/4 \approx 1.96 \times 10^{-7}\ m^2$. $L = RA/\rho = (10 \times 1.96 \times 10^{-7})/(1.6 \times 10^{-8}) \approx 122.7$ m [2]

- Area: 1
- Length ≈ 122.7 m: 1

(ii) The area becomes 4 times, so the resistance becomes one-fourth (2.5 Ω). [1]

Q10. Three resistors of 5 Ω, 10 Ω and 30 Ω are connected in parallel to a 12 V battery. **[3]**

(i) 1/R = 1/5 + 1/10 + 1/30 = 10/30, so R = 3 Ω [1]

(ii) 12/5 = 2.4 A; 12/10 = 1.2 A; 12/30 = 0.4 A [1]

(iii) 2.4 + 1.2 + 0.4 = 4 A (check: 12 V/3 Ω = 4 A) [1]

SECTION D

Q11. Answer the following: **[5]**

(i) The heat produced in a resistor is directly proportional to the square of the current, the resistance, and the time for which the current flows: $H = I^2Rt$. [1]

(ii) $H = I^2Rt = 5^2 \times 20 \times 30 = 15000$ J $= 1.5 \times 10^4$ J [2]

- Formula and substitution: 1
- Answer with unit: 1

(iii) Total power = 100 + 60 = 160 W. I = P/V = 160/220 ≈ 0.73 A [2]

- Total power: 1
- Current ≈ 0.73 A: 1

OR, answer: (a) In parallel, each appliance gets the full supply voltage, and each can be switched on or off independently. If one appliance fails, the others keep working. (b) Energy = 0.4 kW × 8 h × 30 = 96 kWh. Cost = 96 × ₹3.00 = ₹288.

- (a) two reasons: 1 + 1
- (b) energy in kWh 2, cost 1

SECTION E

Q12. In an experiment, a student measured the current (I) through a resistor for different potential differences (V). The readings were: I = 0.5, 1.0,… **[4]**

(i) V is directly proportional to I (Ohm's law). The V–I graph is a straight line through the origin. [1]

(ii) R = V/I = 6.7/2.0 ≈ 3.4 Ω [1]

(iii) Resistance is proportional to length, so R doubles to about 6.7 Ω. I = V/R = 6.7/6.7 ≈ 1.0 A [2]

- R doubles: 1
- I ≈ 1 A: 1